

Government-Grade Smart Water Distribution and Regional Monitoring Platform

Executive assessment

Your idea is much bigger than a normal “water dashboard.” The right way to think about it is as a **Regional Water Operations, Accounting, Monitoring, and Decision-Support Platform**: a system that receives live measurements from rivers and canals, understands the topology of how water splits through the network, compares actual deliveries with approved allocations, detects abnormal distribution, supports operators, preserves an auditable government record, and eventually helps predict shortages and surpluses.

This direction is particularly well aligned with Uzbekistan's current water-sector strategy. In August 2025, the government approved the Water Resources Management and Irrigation Sector Development Program for 2025–2028; official plans include automating **18,755 hydrological stations**, integrating **11 sectoral programs into a single platform**, creating an IT center and situation center, and accumulating pump and water-meter data on a digital platform. ¹ In February 2026, the Ministry of Water Resources also stated that smart water metering will become mandatory for businesses starting in 2027 and that the devices will be integrated into the ministry's **“Water Accounting” information system**. ² In May 2026, the ministry's Digitalization Center announced a procurement for 272 autonomous hardware/software complexes for online measurement of inflows and outflows at irrigation facilities. ³

So the most important strategic recommendation is:

Do not build this as an isolated dashboard. Build it as an interoperable regional water-management platform that can become a module of, or exchange data with, Uzbekistan's national Water Accounting and water-sector digital platforms.

This becomes even more important because a July 2026 World Bank regional program explicitly includes digitization of water accounting and modernization of information systems in the Amu Darya and Syr Darya basins. ⁴

Your proposed five screens are a good starting point, but the current concept is missing several systems that will determine whether it becomes a government-grade product or merely a visualization application:

Area	Current idea	Government-grade version
Dashboard	Charts and hotspots	Regional command center with allocation compliance, water balance, alarms, forecasts and system health
Live information	Device table	Full telemetry and device operations center

Area	Current idea	Government-grade version
Map	83 hotspots	GIS + hydrological network graph + drill-down into every split
Analytics	Bar/line charts	Water accounting, trends, losses, equity, forecasting and anomaly analysis
Reporting	Reports	Legally auditable, versioned and scheduled reporting
Alerts	Basic notifications	Alarm lifecycle, acknowledgment, escalation and incident workflow
Measurement	Depth and m ³	Stage + discharge + cumulative volume + measurement uncertainty
Network	River sections	Formal parent/child hydrological topology
Data quality	Not specified	Calibration, validation, quality flags and provenance
Security	Not specified	Government IAM, OT/IT separation, encryption, audit logs and disaster recovery
Operations	Monitoring	Monitoring first; controlled gate-operation workflow later
Integration	Not specified	APIs, OGC standards, national platforms, GIS and satellite data

My strongest recommendation is to divide the product concept into **three layers**:

Measurement layer → Water Intelligence layer → Government Operations layer.

That separation makes the system scalable, explainable, auditable and significantly easier to procure and maintain.

What the system must actually measure

There is one fundamental technical issue to resolve before UI design or software architecture.

Water depth alone does not reliably tell you how many cubic metres of water are flowing.

Hydrologists distinguish **stage**—the water level—from **discharge**, normally expressed as cubic metres per second. USGS explains that stage does not directly determine flow; a site-specific stage-discharge relationship, commonly called a **rating curve**, has to be established from measurements. The relationship depends on channel geometry and other local conditions and can change when the channel changes. ⁵ ISO 4373:2022 covers instrumentation for measuring water level, while ISO 748:2021 covers methods for determining discharge using velocity and cross-sectional area, including acoustic Doppler techniques. ⁶ ISO 18320:2020 specifically addresses determination of stage-discharge relationships at gauging stations.

⁷

Your system should therefore work with four different concepts:

Stage $h(t)$ = water level in metres

Discharge $Q(t)$ = instantaneous flow in m³/s

$$\text{Volume } V = \int_{t_1}^{t_2} Q(t) dt$$

$$\text{Allocation deviation} = \frac{V_{\text{actual}} - V_{\text{planned}}}{V_{\text{planned}}} \times 100\%$$

That means a dashboard should never casually mix **m³**, **m³/s** and **metres of depth**. They answer completely different operational questions.

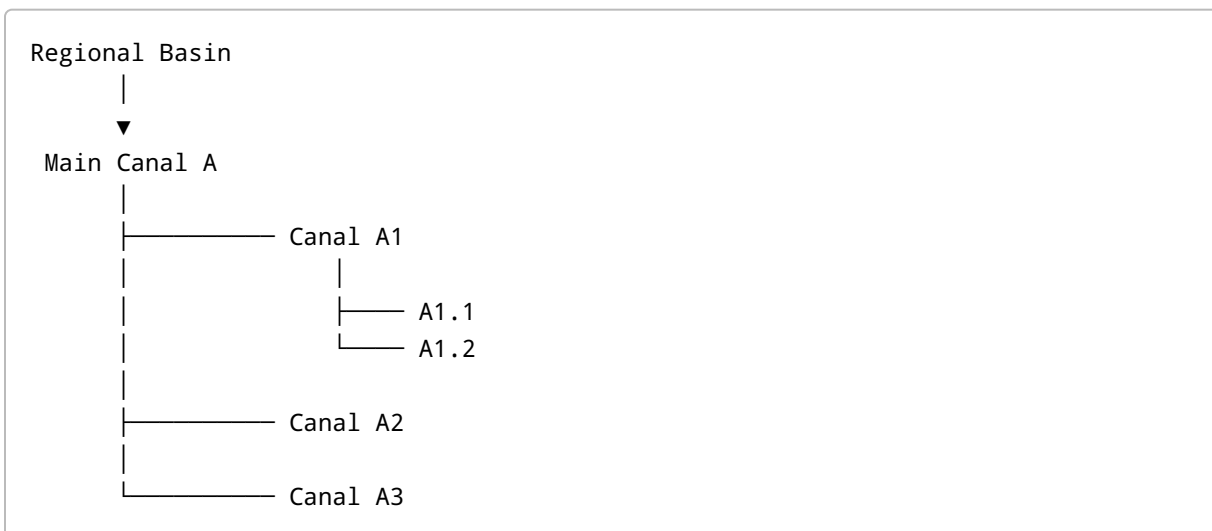
For example:

Canal A water level: **1.82 m**
Current discharge: **12.4 m³/s**
Delivered today: **436,200 m³**
Planned by this time today: **420,000 m³**
Difference: **+16,200 m³ / +3.9%**
Status: **Over allocation**

That one presentation is much more operationally useful than simply showing “water = 1.82 m.”

The river network must become a mathematical graph

For every river or main canal that divides into several canals, your database should know its topology:



Every connection is a **junction**, and every junction should know:

parent waterway
child waterways

```
upstream gauge
downstream gauges
gate/control structure
approved allocation
measurement method
expected conveyance behaviour
travel time
operational owner
administrative area
```

This is critical because your “83 hotspots” are not really 83 isolated map markers. They are **83 entrances into a hydrological network**.

The underlying model should resemble:

```
Basin
↓
River / Main Canal
↓
Junction
↓
Section
↓
Gate
↓
Sub-canal
↓
Measurement Station
↓
Device / Sensor
↓
Observation
```

OGC's WaterML standards provide standardized models for hydrological features, measurements, ratings, gaugings and channel sections, while the OGC SensorThings API provides a standardized geospatial interface for IoT observations and sensor metadata. Using those concepts will make future integration much easier than inventing an entirely proprietary sensor model. ⁸

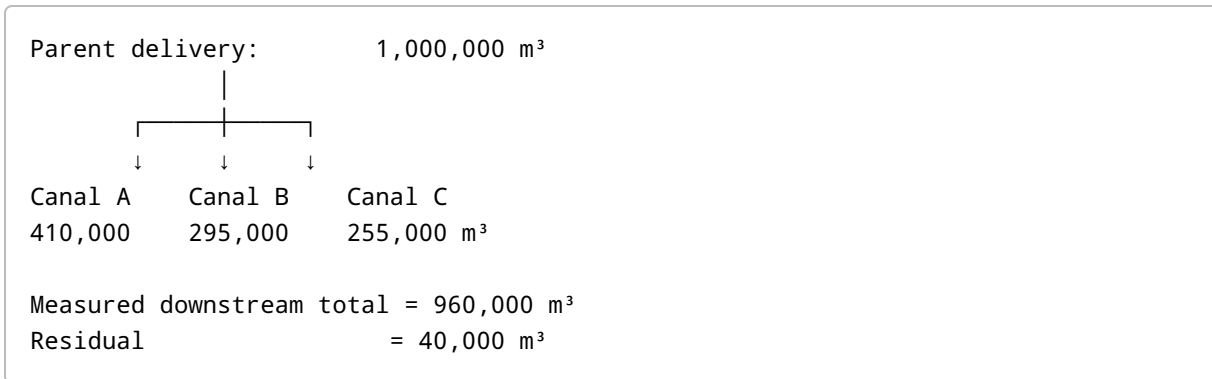
Water balance is the core intelligence

At a split, a simplified accounting calculation is:

$$R = V_{\text{parent}} - \sum V_{\text{children}} - \Delta S - \text{estimated legitimate losses/returns}$$

where R is the unexplained residual and ΔS accounts for changing storage.

Suppose a main canal supplies:



The platform should ask whether that 40,000 m³ represents expected conveyance/storage behaviour, measurement uncertainty, an unmetered withdrawal, a gate problem, data quality failure, or something requiring investigation.

This is much more powerful than simply saying one section is red.

Water accounting and auditing is specifically recommended by FAO as a foundation for dealing with water scarcity and improving water-management decisions; its framework considers water availability, use, infrastructure and governance rather than treating individual sensor readings in isolation. ⁹ The World Bank similarly identifies adequacy, reliability, equity and productivity among useful irrigation performance dimensions. ¹⁰

One complication your engineers must not overlook is **travel time**. If water takes two hours to travel between an upstream measurement point and downstream branches, comparing all stations at exactly 10:00 can produce a false imbalance. The analytics engine therefore needs time-shifting or hydraulic routing where relevant. This is an engineering inference from the fact that the application is comparing spatially separated measurements rather than observations made at a single cross-section.

Every number needs a confidence level

Government staff should never be shown a derived value as though every measurement has perfect accuracy.

Each observation should carry:

```
value
unit
timestamp
device
measurement method
quality flag
uncertainty/confidence
```

calibration status
raw/validated/corrected state

USGS notes that rating relationships can change because debris, vegetation, floods and changes in channel geometry affect the stage-discharge relationship, requiring periodic verification. ¹¹ WMO likewise treats the design, operation and data-quality practices of hydrological monitoring networks as an end-to-end process rather than merely installing sensors. ¹²

I would therefore use four data states:

Raw → Automatically validated → Expert validated → Corrected/estimated

Never silently overwrite historical measurements. Corrections should create a new revision with the original retained.

That feature will become extremely important when government officials later ask:

“Why does the annual report show 41.6 million m³ today when last month's report showed 40.9 million?”

The system must be able to answer exactly which observations were recalibrated, corrected or estimated.

Recommended product structure

I would keep your five-page idea. Five pages is enough if each page is designed around a clear operational job rather than simply around a visualization type.

Proposed application structure

Page	Primary user question
Command Dashboard	What requires my attention right now?
Live Operations	What exactly are all stations and devices reporting?
Water Network Map	Where is water going and which branches are abnormal?
Analytics & Forecasting	Why is it happening, what has changed, and what is likely next?
Reports & Audit	What must be documented, approved and reported?

Alerts should not become a separate sixth page. Put an **Alarm Center available globally from every screen**, with a dedicated side panel or full-screen mode.

Command Dashboard

The dashboard should not primarily be a collection of charts. It should answer an operator's questions in approximately five seconds:

- How much water do we have?
- How much should have been delivered?
- Where are we over or under plan?
- Is anything dangerous?
- Are our measurements trustworthy?

The first row should contain six or seven high-value KPIs:

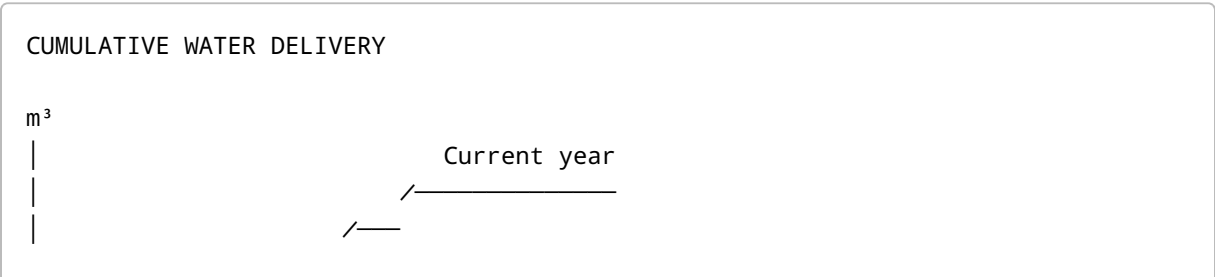
KPI	Example
Regional inflow	18.4M m³ today
Delivered	17.1M m³
Planned delivery	16.8M m³
Allocation compliance	92.7%
Unexplained balance	1.8%
Active critical alarms	7
Stations online	97.6%

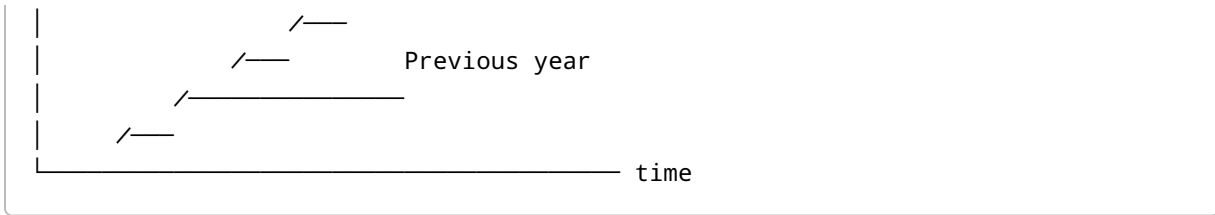
Immediately below it, I would place a **regional situation map** rather than another KPI row.

Then show:

MOST IMPORTANT DEVIATIONS			
Canal A7	+18.4%	+83,200 m³	42 min
Canal D3	-16.1%	-51,900 m³	1h 14m
Canal F1	+12.8%	+31,400 m³	18 min

Next comes your year-over-year chart:





But compare the **same point of the irrigation/water-management season**, not just arbitrary calendar dates. Also show plan alongside current-year and previous-year values so that officials do not incorrectly conclude that “more than last year” necessarily means “bad.”

A useful second chart would show:

Planned vs Actual vs Previous Year

and allow switching between:

Today | Week | Month | Irrigation season | Year

The dashboard should also show **system confidence**, because a perfect-looking water-distribution chart is misleading when 15% of its underlying sensors have not reported for two hours.

Live Operations

The live page should become the operational telemetry center.

I recommend columns such as:

Field	Example
Station	FERG-0217
Device	DEV-92827
Waterway	Katta Fergana Canal
Section	KF-04
Stage	1.84 m
Flow	11.8 m ³ /s
Today volume	482,100 m ³
Planned	460,000 m ³
Variance	+4.8%
Quality	Valid
Status	Over

Field	Example
Last update	8 sec ago
Power	Normal
Signal	82%
Calibration	Valid
Alarm	Warning

A government operator needs significantly more than your proposed device number and reading.

Filters should cover:

Region → Basin → Waterway → Section → Station → Device type → Status → Data quality

Clicking a row should open a **device/station inspector** without navigating away.

That inspector should show the last 24-hour graph, raw and validated observations, communication health, power/battery status, sensor metadata, installation date, last calibration, next calibration, firmware, recent alarms, maintenance history and photos/documents for the physical installation.

This is where your project evolves from a monitoring website into an **asset-management system**.

Water Network Map

This has the potential to become the most impressive and most useful page in the entire project.

But I would change the concept of “83 giant hotspots.”

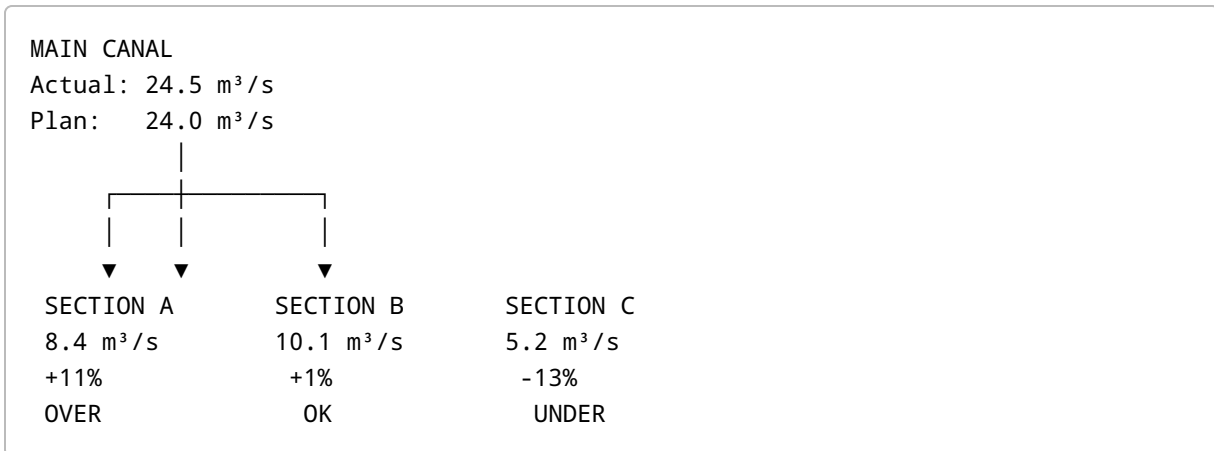
At regional zoom, 83 giant pins will produce visual clutter. Instead show basin/river-level status and progressively reveal junctions as the operator zooms in.

For example:

REGIONAL VIEW

- Amu-Kashkadarya 3 critical
- Lower Zarafshan normal
- Chirchik-Ohangaron 2 warnings
- Syrdarya-Sokh 5 critical

Clicking one opens its network.



The real map and the network diagram should complement one another:

Geographic view tells officials *where* something is.

Hydrological network view tells them *how the water is connected*.

On selecting a branch, open a side panel containing:

Section: Canal B-17

Current discharge:	8.74 m ³ /s
Target discharge:	8.20 m ³ /s
Difference:	+0.54 m ³ /s
Delivered today:	524,310 m ³
Planned today:	491,000 m ³
Difference:	+33,310 m ³ (+6.8%)
Deviation duration:	47 minutes
Sensor confidence:	High
Last observation:	12 seconds ago
Gate position:	63%
Responsible office:	District Water Service

I strongly agree with your red/green/yellow concept, but it must not depend on color alone. WCAG explicitly requires that color not be the sole means of conveying information, and government digital-service guidance increasingly targets WCAG 2.2 AA accessibility. ¹³

Use:

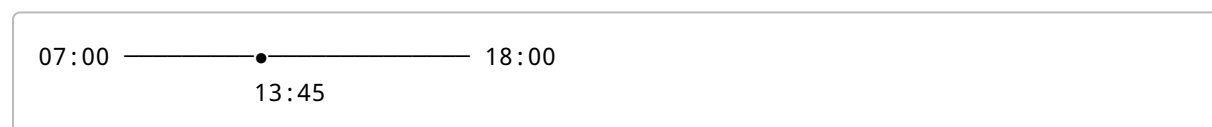
State	Visual	Meaning
▲ OVER	Red	More than allowed

State	Visual	Meaning
✓ ON PLAN	Green	Within tolerance
▼ UNDER	Yellow/amber	Less than allocation
? NO DATA	Grey	Measurement unavailable
⚠ DEVICE FAULT	Grey/hatched/icon	Data is unreliable

This fourth and fifth state are important. A disconnected sensor must never look like “everything is okay.”

The map should eventually support layers for hydraulic structures, gates, pumping stations, reservoirs, satellite information, irrigation areas and administrative boundaries. Uzbekistan is already expanding satellite-supported water monitoring: Uzbekspace stated in May 2026 that space monitoring can estimate actual agricultural water consumption and support a “virtual water meter” approach. ¹⁴ Satellite data should be treated as a complementary validation layer, not as a replacement for hydrometric measurements at critical distribution structures.

A **time-travel slider** would make this page exceptionally powerful:



Operators could replay the distribution network and see exactly how a shortage or surplus propagated through canals during an incident.

Analytics and Forecasting

I would not make this page simply “bar and line charts.” Instead, organize it around questions.

Delivery performance

Show planned versus actual delivery by basin, canal, district and water user.

Distribution equity

Highlight where one branch systematically gets more than its allocation while another consistently receives less. The World Bank’s irrigation work identifies reliability, adequacy and equity as important service-performance dimensions, so this is directly relevant to a public-sector irrigation platform. ¹⁵

Water balance

Use a waterfall chart:

Main intake	10.00M m ³	
Measured branch deliveries	-9.31M m ³	
Conveyance/storage estimate	-0.42M m ³	
Unexplained	0.27M m ³	

Deviation heatmap

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Canal A	✓	✓	▲	▲	✓	✓	✓
Canal B	▼	▼	▼	✓	✓	▼	▼
Canal C	✓	✓	✓	✓	✓	✓	✓
Canal D	▲	▲	▲	▲	▲	✓	✓

An official can immediately recognize systematic distribution problems.

Historical analysis

Allow current year versus previous year, but add:

- planned allocation,
- cumulative actual,
- percentage variance,
- previous-period average,
- maximum/minimum,
- weather/season context where available.

Forecasting

This should become a second-stage feature.

Useful forecasts include predicted inflow for 24–72 hours, likely allocation shortfall, sections likely to exceed plan, sensor failure probability, demand forecast and expected reservoir/canal conditions.

I would initially keep the AI **advisory rather than autonomous**:

“Section B is likely to fall 11% below today's allocation by 18:00.”

rather than:

"AI automatically opens gate B."

That separation matters because this platform influences physical infrastructure and allocation of a strategic resource. Uzbekistan's AI strategy emphasizes regulatory frameworks, evaluation and transparent implementation of AI projects. ¹⁶

Later, the application can add a **water-distribution digital twin**:

"What happens if the upstream flow falls 12%?"

System simulation:

Canal A	-4%	
Canal B	-8%	
Canal C	-17%	▲
Canal D	-23%	▲ CRITICAL

That turns the platform from monitoring into genuine decision support.

Reports and Audit

Reporting should be considered a major subsystem, not merely "export chart to PDF."

I recommend at least these standard government reports:

Report	Frequency
Regional Water Situation	Daily
Allocation Compliance	Daily/weekly
Canal Water Balance	Daily/monthly
Delivery by district	Weekly/monthly
Over/under allocation incidents	On demand/monthly
Station availability	Weekly
Device and communication health	Weekly
Calibration compliance	Monthly
Seasonal Water Balance	Seasonal
Year-over-Year Water Use	Annual
Alarm/Incident Report	Per incident
Management Executive Report	Monthly

Report	Frequency
Audit and Data Correction Report	Monthly/on demand

Every report should record:

```

Report version
Reporting period
Data cut-off timestamp
Data revision/snapshot
Calculation methodology
Quality status
Generated by
Approved by
Generation time

```

The user should be able to reproduce a six-month-old report using the exact version of the data and calculation rules that existed at the time.

FAO's water accounting and auditing guidance specifically treats accounting as input to decision-making and governance, rather than a one-time visualization exercise. ¹⁷

Operational logic and alerting

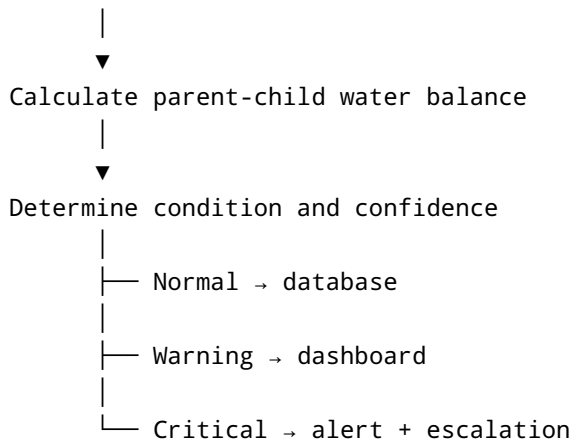
The biggest software intelligence in this project should actually sit **between the sensors and the UI**.

A basic rule engine might look like:

```

Observation
  |
  ▼
Quality validation
  |
  ▼
Convert stage → discharge where required
  |
  ▼
Integrate discharge → volume
  |
  ▼
Apply network / travel-time logic
  |
  ▼
Compare actual with approved allocation

```



Do not trigger alarms from one bad measurement

Suppose a sensor briefly reports:

```
09:00  8.1 m³/s
09:01  8.2
09:02  57.0 ← erroneous spike
09:03  8.3
```

A naive system sends a critical government alert.

A proper system checks physical plausibility, sensor quality, rate-of-change, adjacent stations and persistence.

Instead of:

```
IF actual > planned
  SEND ALERT
```

use rules closer to:

```
IF
  quality = valid
  AND deviation exceeds configured tolerance
  AND condition persists for configured duration
THEN
  create operational alarm
```

Add **hysteresis**, so an alarm does not constantly alternate between normal and warning when a measurement moves around the threshold.

The exact tolerances should not be hard-coded globally. They should be configurable by canal, season, operating policy, measurement uncertainty and allocation agreement.

Use alarm severity, not just status

Water status and alarm severity are different concepts.

A canal could be **8% under plan** but only an advisory early in the day, while a 3% shortfall at a critical strategic intake might require immediate intervention.

I recommend:

INFORMATION

No action required.

ADVISORY

Operator should observe.

WARNING

Intervention likely required.

CRITICAL

Immediate action/escalation required.

ISA-18.2 defines a lifecycle approach to industrial alarm management including identification, rationalization, prioritization, implementation, monitoring and auditing; its goal includes avoiding alarm overload and helping operators maintain situational awareness. ¹⁸ That philosophy fits your water-management platform very well.

Every alarm should have:

Severity

Location

Cause

Actual value

Expected value

Deviation

Start time

Duration

Suggested procedure

Assigned organization

Acknowledged by

Acknowledged at

Comment

Resolution
Closed by

Then your system knows the difference between an **anomaly** and an **incident**.

A realistic workflow would be:

```
Water deviation detected
↓
Warning created
↓
Regional operator acknowledges
↓
Operator investigates
↓
Maintenance / field team assigned if necessary
↓
Corrective action recorded
↓
Flow returns to normal
↓
Alarm cleared
↓
Incident closed
↓
Included automatically in monthly report
```

Add alarms that are not about allocation

A government platform should detect far more than excess or insufficient water.

Important event classes include:

Event	Example
Over-allocation	+14% for 30 minutes
Under-allocation	-18%
Unexplained balance	Parent/children discrepancy
Sudden flow change	-40% in 4 minutes
High stage	Flood/overflow concern
Dry canal	Unexpected zero flow

Event	Example
Sensor frozen	Same reading for excessive duration
Impossible measurement	Physical limit exceeded
Communication loss	No telemetry
Power problem	Battery/power failure
Calibration overdue	Measurement confidence reduced
Gate mismatch	Command/position inconsistency
Network inconsistency	Upstream/downstream values incompatible

This gives you two parallel statuses:

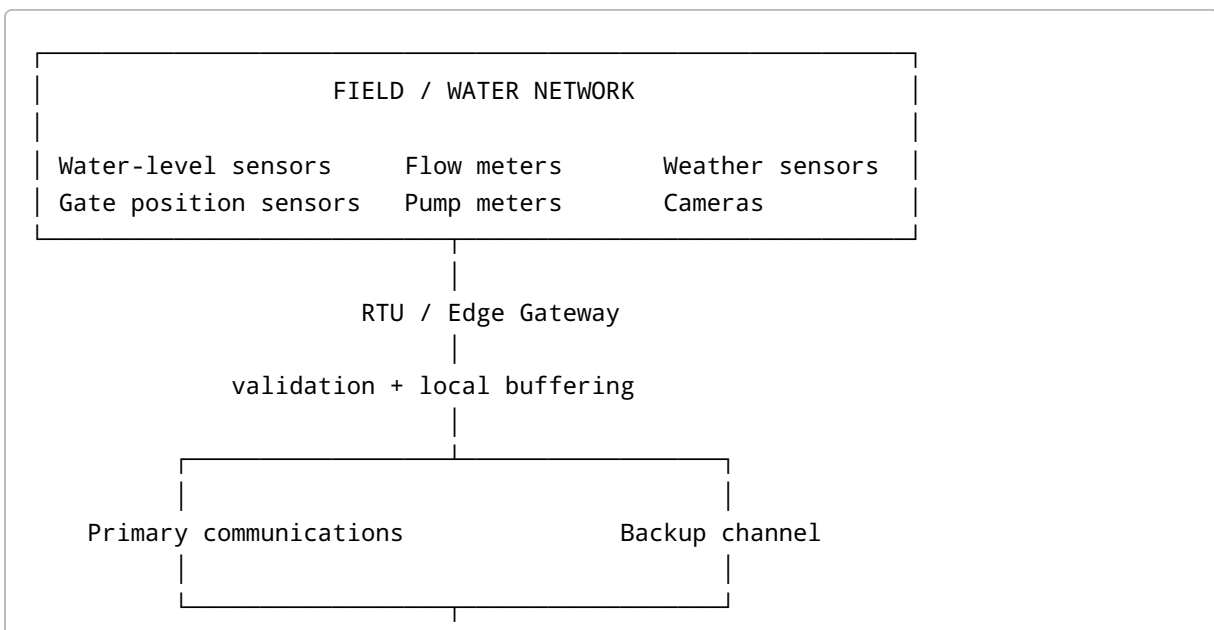
Water condition and **System/device condition**.

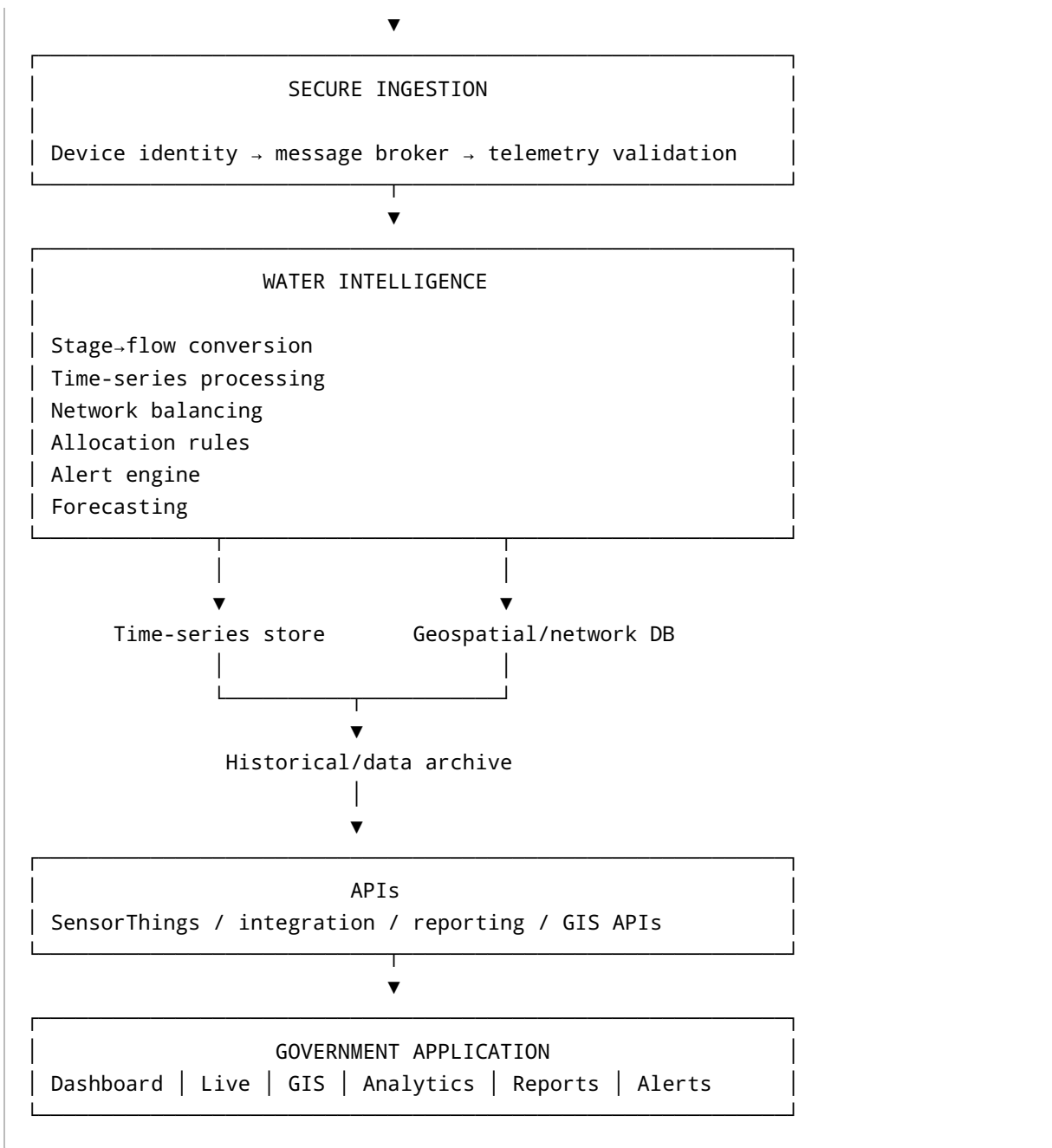
That distinction will save operators from a major problem: treating bad instrumentation as a real hydraulic emergency.

Government-grade architecture

The architecture should be designed for substantially more than your initial 83 hotspot locations. Uzbekistan's national strategy itself demonstrates that the surrounding ecosystem can involve tens of thousands of monitoring points, so vendor-neutral scalability and interoperability are important from the beginning. ¹

A strong conceptual architecture would look like this:





OGC SensorThings is particularly useful as an interoperability model because it provides a standardized, geospatially enabled way to manage observations and metadata across heterogeneous IoT systems. ¹⁹ WaterML should be considered for hydrological exchange with other agencies and platforms. ²⁰

Edge computing is important

Every critical station should be able to continue recording observations when communications are unavailable.

The device or RTU should:

```
Measure
Timestamp
Validate basic limits
Store locally
Transmit
Retry when disconnected
Synchronize missing observations later
```

Otherwise a cellular outage creates permanent gaps in the government's water record.

For strategically important structures, consider redundant power and communications according to site criticality rather than installing the same configuration everywhere.

Separate OT from IT

This project becomes substantially more security-sensitive the moment it can manipulate gates, pumps or valves.

The architecture should distinguish:

```
Telemetry
  ↓
Operational Technology network
  ↓
Controlled security boundary
  ↓
Government application / IT network
```

Do not expose PLCs, RTUs or gate controllers directly to the public application network.

NIST SP 800-82 Rev. 3 specifically addresses operational technology systems that interact with physical processes and emphasizes that OT security must preserve safety, reliability and availability as well as conventional cybersecurity. ²¹ The IEC 62443 family similarly addresses cybersecurity across the lifecycle of industrial automation and control systems. ²²

This is not theoretical. On July 30, 2026, CISA warned of increased threat activity targeting programmable logic controllers in water and wastewater systems. ²³

For that reason, I would make version one of your platform primarily **monitoring and decision support**.

A later control version can add:

```
Operator requests gate change
↓
System validates limits
↓
Second authorized operator approves
↓
Command sent through OT control layer
↓
PLC confirms execution
↓
Physical position independently verified
↓
Action written to immutable audit history
```

For major water-control decisions, a **four-eyes principle** is much safer than one-button remote control.

Use a proper historical data architecture

You will eventually hold several fundamentally different types of data.

Real-time time-series data

```
time
device
stage
flow
gate_position
battery
signal
quality
```

Geospatial/network data

```
river
canal
section
junction
polygon
gate
station
geometry
topology
```

Business/governance data

allocation plans
users
organizations
incidents
approvals
reports
maintenance records

Large files

photos
inspection documents
calibration certificates
maps
satellite layers
generated reports

Do not try to force all of these into a single database abstraction.

The allocation plan deserves its own engine

This is one of the most important additions I would make.

There should be an **Allocation Management module** behind the five screens.

Example:

Waterway: Canal B
Period: 15 Aug - 31 Aug

Day	Planned volume
15 Aug	480,000 m ³
16 Aug	500,000
17 Aug	520,000
...	

Or time-based:

00:00-06:00	5.0 m ³ /s
06:00-12:00	8.2 m ³ /s
12:00-18:00	9.4 m ³ /s
18:00-24:00	6.7 m ³ /s

Plans should be version-controlled.

Changing an approved allocation should record:

```
previous value
new value
reason
requested by
approved by
effective time
legal/administrative reference
```

Otherwise an administrator could simply change the planned amount after excessive water has already been delivered, making historical compliance meaningless.

That auditability is one of the differences between commercial dashboard software and a government resource-management system.

Security, governance, and standards

Because this system would manage a strategically important physical resource, I would treat security and governance as first-class product requirements rather than a final “security phase.”

Uzbekistan's Law “On Cybersecurity,” effective since July 2022, establishes the national framework for cybersecurity and contains specific provisions concerning critical information infrastructure, cybersecurity incidents and state oversight. Whether this particular regional platform is formally classified as critical information infrastructure would need to be established with the responsible authorities, but that assessment should happen near the beginning of the project rather than after deployment. 24

NIST Cybersecurity Framework 2.0 organizes cybersecurity around six functions—**Govern, Identify, Protect, Detect, Respond and Recover**—which is an excellent structure for the project's security requirements. 25

For your platform this translates into:

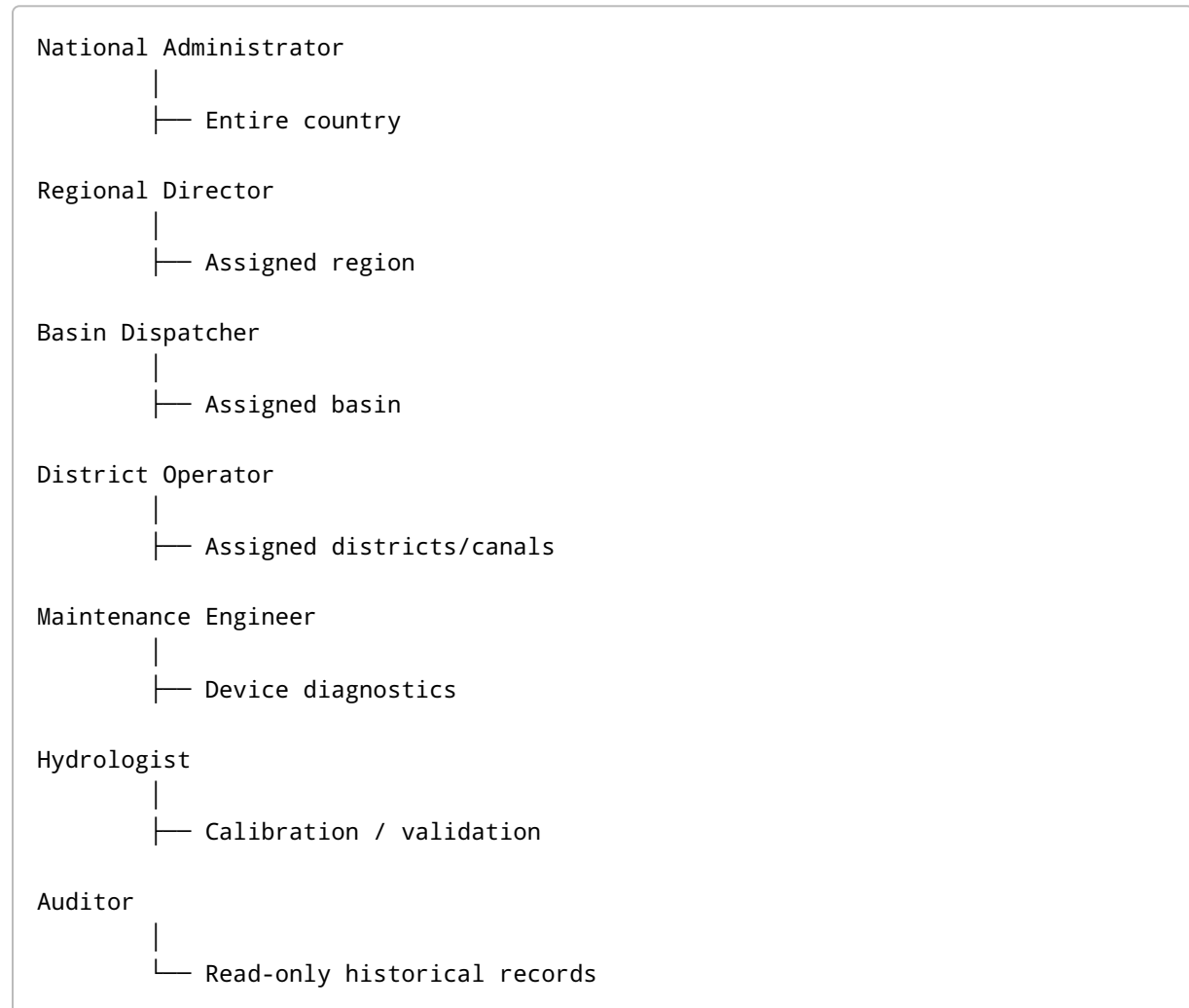
Function	Water-platform implementation
Govern	Security ownership, policies, supplier requirements
Identify	Complete inventory of sensors, RTUs, servers and users
Protect	MFA, encryption, segmentation, least privilege
Detect	Security monitoring and anomaly detection
Respond	Cyber incident procedures and escalation

Function	Water-platform implementation
Recover	Backups, failover, restoration and manual operations

User roles must follow geography and responsibility

I would build access around both role and territory.

Example:



A district operator should not be able to modify another region's allocations simply because both are technically "operators."

Sensitive operations should require multi-factor authentication, and privileged accounts should be strictly separated from normal accounts.

Everything important should be audited

Log at least:

- Login
- Logout
- Failed authentication
- Data export
- Report generation
- Allocation modification
- Threshold modification
- Alarm acknowledgment
- Alarm closure
- Device configuration
- Calibration modification
- Manual data correction
- Control command
- Permission change

For each:

- Who?
- What?
- When?
- Where?
- Old value?
- New value?
- Reason?

That protects not only against cyberattack but also against disputes and corruption.

Procurement must avoid vendor lock-in

A major government project can last considerably longer than an individual sensor supplier or software contract.

The state should own:

the data model, historical data, GIS topology, API specification, alert definitions, allocation records and export formats.

Sensors and dashboards should be replaceable without losing the system.

OGC SensorThings and WaterML are particularly useful for this objective because they provide open interoperability models for sensor and water information rather than locking the project into a proprietary representation. ²⁶

The procurement specification should also require full API documentation, bulk historical export, database/ data dictionary documentation, configuration export, no undocumented proprietary encryption, source/ configuration escrow where appropriate, and clearly defined ownership of all government-generated data.

Accessibility and languages

For government deployment in Uzbekistan, I would design the application from day one for **Uzbek, Russian and English** rather than hard-coding English text into the frontend.

Also target WCAG 2.2 AA. W3C specifically requires that information not rely solely on color and defines contrast requirements for accessibility. ²⁷ Government digital-service guidance such as the UK Service Standard likewise requires accessible services and user research rather than treating accessibility as cosmetic compliance. ²⁸

Therefore even your red/yellow/green system should always display:

▲ OVER +12%
✓ ON PLAN
▼ UNDER -9%
? NO DATA

not merely colored circles.

Delivery roadmap and definition of a “perfect” version

I would **not** try to build the complete intelligent control platform in one release. Government infrastructure projects become risky when sensing, analytics, GIS, reporting, AI and remote control are introduced simultaneously before the underlying measurements have been proven trustworthy.

A much safer maturity path is:

Stage	Capability
Foundation	Water-network topology, station registry, users, GIS
Monitoring	Real-time telemetry and historical database
Accounting	Stage/flow/volume, allocation comparison, water balance
Operations	Alarm workflows, device health, incident management
Governance	Reports, audit logs, approvals, calibration records

Stage	Capability
Intelligence	Forecasts, anomaly detection, satellite integration
Simulation	Network/digital-twin scenarios
Control	Carefully governed remote gate/pump operation

The first production milestone should therefore prove **measurement integrity and water accounting**, not AI.

A strong pilot could cover:

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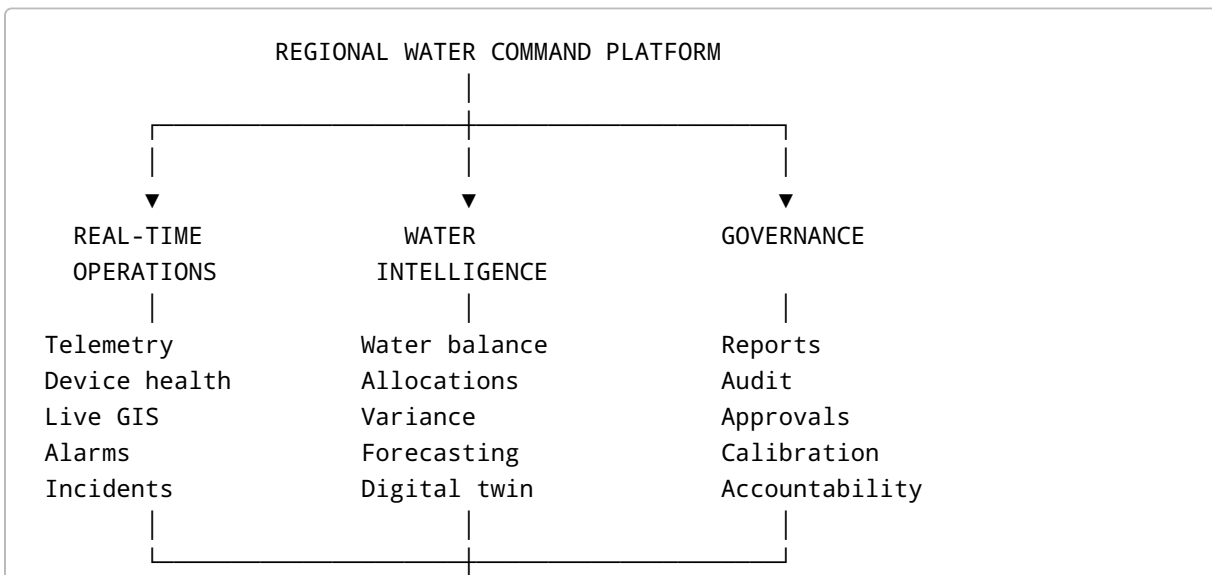
1 major river/canal
  ↓
3-5 major distribution junctions
  ↓
15-30 monitored branches
  ↓
full allocation + telemetry + alarm + report workflow

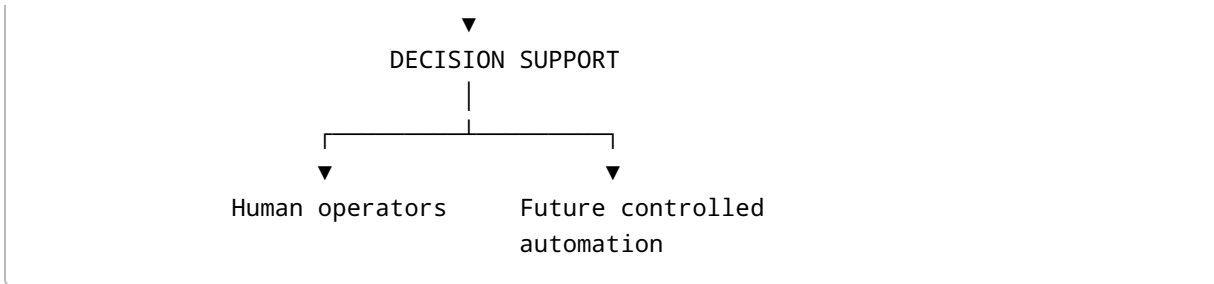
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Run that through an irrigation period and discover the real-world problems—sensor drift, missing telemetry, rating changes, communication loss, administrative-plan changes and operator workflows—before expanding across all 83 hotspots.

The World Bank's irrigation modernization work stresses that infrastructure and associated management systems have to be assessed together; performance systems should systematically observe, document and analyze predefined indicators rather than concentrating only on hardware installation. ²⁹

The full government product I would ultimately specify looks like this:





To judge whether it is genuinely ready for government operation, I would put the following items directly into the acceptance criteria rather than leaving them as optional future improvements:

Acceptance area	Government-level expectation
Hydrometry	Every important station has documented measurement methodology
Calibration	Calibration status and history are available online
Data quality	Every observation has quality/provenance status
Availability	Missing telemetry is clearly detected
Water accounting	Parent/child balance is calculated correctly
Allocation	Plans are versioned and approved
GIS	Every monitored asset has formal spatial/network identity
Alerts	Alerts are prioritized, acknowledged and auditable
Reporting	Historical reports are reproducible
Security	OT/IT segregation, MFA, logging and incident procedures
Resilience	Backup, disaster recovery and field buffering
Interoperability	Documented APIs and open water/sensor formats
Accessibility	WCAG-aligned UI; color is not the only status indicator
Localization	Uzbek/Russian/English-ready architecture
Auditability	Administrative and operational changes are traceable
Scalability	Architecture accommodates far more than the initial 83 sites
Ownership	Government owns data, definitions and export capability

The biggest conceptual upgrade is therefore this:

Do not build a system that tells officials “this canal is red.” Build a system that can prove why it is red, how much water is involved, when the deviation began, how trustworthy the measurement is, where that water came from, which downstream

sections are affected, who is responsible, what action was taken, and what will happen if nothing changes.

That is the difference between a visually impressive dashboard and a **government-level regional water management system**.

And in the Uzbekistan context, there is a particularly strong opportunity: the government's current direction toward a unified water digital platform, automated hydrological stations, smart water accounting, satellite monitoring and situation-center operations means your project should be designed not merely as a frontend for 83 points, but as a **regional operational layer of the emerging national digital water-management ecosystem**. ³⁰

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